International Journal of Hydrogen Energy 114 (2025) 295–311

Contents lists available at ScienceDirect

International Journal of Hydrogen Energy

journal homepage: www.elsevier.com/locate/he

Review on thermal design and thermal management for metal hydride reactors: Current status and future development

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ARTICLE INFO

Handling Editor: Dr A Bhatnagar

Keywords:
Metal hydride reactors
Hydrogen absorption-desorption performance
Thermal design
Coupling thermal management
Machine learning
Heat transfer

ABSTRACT

Metal hydride (MH) hydrogen storage is one of the most important hydrogen storage methods. In MH reactor, the hydrogen absorption/desorption rates are limited by the heat transfer performance, the thermal design of MH reactor is thus particularly important which facilitates rapid heat exchange. Besides, accumulating the absorption reaction heat and supplying it for hydrogen desorption by thermal management design is also of great importance on improving the energy efficiency of the hydrogen storage system. In this review, the current research studies of MH reactors is summarized with emphasis on thermal design and thermal management, and the advantages and disadvantages of different heat transfer structures are analyzed and compared in detail. Specifically, the effect of the thermal design of MH reactors on the gravimetric and volumetric hydrogen storage density is considered, and the hydrogen storage performance indicators for MH reactors are discussed at length. More importantly, the novel point of this review is the prospect of the future development trend of MH reactor thermal design and thermal management, which is the combination of MH reactor and intelligent algorithms and the multiple coupling of MH reactor with other systems such as fuel cells.

1. Introduction

Energy is the cornerstone of the development and survival of modern society. Fossil energy plays a vital role in human energy history, but fossil energy is non-renewable and it has brought a lot of pollution to the environment [1–3]. Therefore, finding and developing green, non-pollution and renewable energy have received enormous attention from researchers and governments around the world [4–9]. Hydrogen energy is regarded as one of the most promising future energy because of its characteristics of green, high-energy density, renewable, non-pollution and wide use [10–13].

Although the calorific value of hydrogen energy is the highest at the same weight, its energy density is relatively low at normal temperature and pressure due to the low density. Thus, hydrogen is generally compressed and filled into a high-pressure hydrogen storage tank to be used, which increases the difficulty of hydrogen storage and also increases the risk. Normally, liquid hydrogen is stored below −253.15 °C and consumes a lot of energy in the process of maintaining low temperature and requires relatively high-quality hydrogen storage tanks. Some scholars have also proposed organic liquid hydrogen storage and underground hydrogen storage. Currently, the development of organic liquid hydrogen storage is not mature, and underground hydrogen storage has high environmental and geological requirements and cannot be widely used [14–17]. In contrast, solid-state hydrogen storage stores hydrogen in solid materials by chemical adsorption or physical adsorption. It has the advantages of high hydrogen storage density per unit volume, the reversible and stable absorption/desorption process, safe, conducive to long-term storage and long-distance safe transport of hydrogen, etc. Therefore, solid-state hydrogen storage can effectively overcome the shortcomings of gas and liquid storage methods.

Solid-state hydrogen storage materials mainly include MH [18], carbon-based material (C_bM) [19], metal-organic skeleton compounds (MSC) [20], complex material (CM) [21] and so on. Among them, MH alloys are adopted extensively due to the most mature development. Metal hydrogen storage alloys include $LaNi_5$ [22–27], FeTi [28–31], Mg [32–35], ZrCo [36–39], DU [40,41] etc. MH is favored by researchers due to it has advantages, such as low pressure of hydrogen storage, high safety, high volume density and recyclability. MH reactors are used in

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https://doi.org/10.1016/j.ijhydene.2025.03.038
Received 9 September 2024; Received in revised form 4 January 2025; Accepted 3 March 2025
Available online 9 March 2025

the absorption/desorption hydrogen process of these metal compounds and are accompanied by a large amount of heat transfer in this process, so many researchers have conducted a lot of research around MH reactor and heat transfer in MH reactor.

In this review, as shown in Fig. 1, firstly, the key thermal problem of MH reactor is analyzed, because it must be paid attention to by researchers. Secondly, the thermal design and thermal management of MH reactor are summarized respectively. The thermal design always includes adding high thermal conductivity materials, fins and heat exchange tubes. The thermal management includes that MH reactor is combined with phase change material (PCM), thermo-chemical heat storage (THS) and MH heat pump for coupling. Then, the comprehensive hydrogen storage performance (HSP) standard for the design of MH reactor is summarized. Finally, since intelligent algorithms such as machine learning have been gradually applied to the thermal management and thermal design of MH reactor, this review is novel to summarize in summarizing and prospect prospecting the development prospects of the hydrogen reactor, including how the hydrogen reactor can be combined with fuel cells (FC) and combined heat and power (CHP), and how they and MH reactor can achieve good heat utilization.

2. The key thermal issues in MH reactors

Metal hydride hydrogen storage is that some metals or metal alloys can undergo a combined adsorption reaction with hydrogen (reaction formula as indicated in Eq. (1)), and the reaction process is reversible.

$$\mathrm{M} + \frac{x}{2}\mathrm{H_2} \leftrightarrow \mathrm{MH_x} + \Delta H \tag{1}$$

where M is metal or metal alloy, and MH_x is the metal hydride. The above reaction has a significant thermal effect, and the process of hydrogen absorption reaction is accompanied by heat release, and the process of hydrogen discharge reaction is accompanied by heat absorption. If the heat emitted in the hydrogen absorption reaction cannot be heat transfer in time, it will affect the normal progress of the reaction. On the contrary, the heat cannot be quickly supplemented in desorption hydrogen process, and the hydrogen cannot be continuously and steadily released.

The thermodynamic characteristics of the MH alloys can be described by the pressure-composition-temperature (P–C-T) curve, as shown in Fig. 2. In the first stage (O-A) of the hydrogenation process, hydrogen dissolves into the metal to form α solid solutions, and the

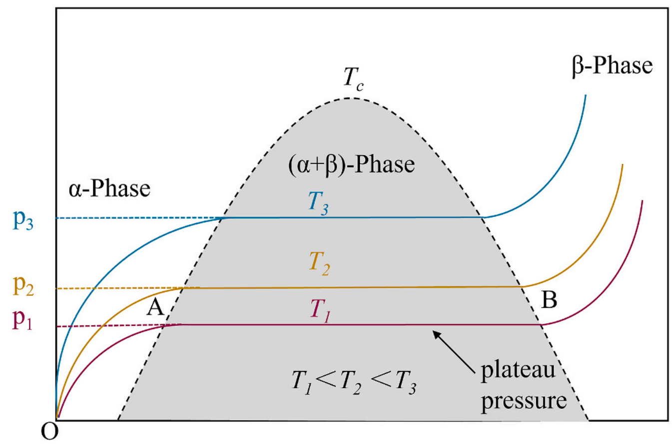

Fig. 2. Diagram of P–C-T curve of MH absorption/desorption of hydrogen.

equilibrium pressure increases sharply. With the reaction progressing, the α solid solution gradually changes from α-phase to metallic β-phase, with the pressure remaining constant. In this process (A-B), the MH alloy will absorb a large amount of hydrogen, and the pressure is called "plateau pressure", which is an important parameter indicating the stable hydrogen absorption/desorption pressure of the MH alloy and thus guiding the choose of alloy for different hydrogen storage systems. The width of "plateau pressure" reflects the hydrogen storage capacity of MH alloy. When the temperature rises, the width will narrow down, indicating that the hydrogen storage capacity of MH is weakened and the temperature rise has a negative influence on the hydrogen storage capacity. According to the principle of chemical equilibrium, the relationship between the temperature of hydrogen absorption and desorption in P–C-T curve and the corresponding plateau pressure can be expressed by the Van't Hoff equation:

$$\ln p_{eq} = \frac{\Delta H}{RT} - \frac{\Delta S}{R} \tag{2}$$

where ΔH, R, T and ΔS are the reaction enthalpy, universal gas constant, temperature and reaction entropy, respectively, p_{eq} is the thermodynamic equilibrium pressure.

The increase of equilibrium pressure will further directly affect the hydrogen absorption capacity. In solid hydrogen storage, increasing the amount of alloy can increase the amount of hydrogen storage, but it

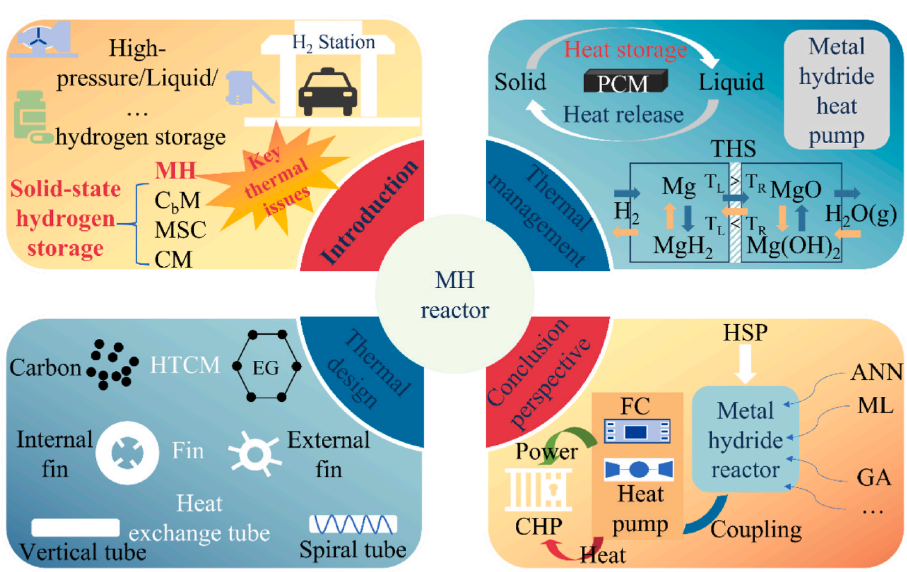

Fig. 1. Table of content diagram of introduction, thermal design, thermal management, conclusion and perspective.

cannot solve the problem of hydrogen storage reaction rate. Shafiee et al. [42] through summarizing and analyzing the absorption/desorption hydrogen performance of various MH reactors, considered that the absorption/desorption hydrogen rate in the MH reactors was usually limited by the heat transfer rate, rather than the intrinsic kinetic performance of hydrogen storage alloys. Thus, heat transfer enhancement is needed to improve the heat transfer performance in MH reactors. For further analysis of influencing factors, we can see Eq. (3).

$$\dot{m} = -C_a \ln\left(\frac{p_g}{p_{eq}}\right) \exp\left(\frac{-E_a}{RT}\right)(\rho_{s,sat} - \rho_s) \quad (3)$$

where $\dot{m}$ is the reaction rate per unit volume of hydrogen, C_a is absorption rate constant, ρ_g is density of hydrogen gas, E_a is activation energy-absorption, $\rho_{s,sat}$ and ρ_s are saturated density and density respectively [43].

After analyzing the kinetics of hydrogen absorption reaction, it can be concluded that another factor affecting the reaction rate is temperature. Temperature is another important factor affecting the hydrogen absorption/desorption rate, which will also guide us to focus on reducing the reaction temperature. This means that the heat should transfer in time, which requires the MH reactor to conduct heat as quickly as possible. In other words, we should try our best to solve the key thermal issues in MH reactors, so that the heat transfer of MH reactors can be made faster so that the reaction can be carried out more smoothly.

During hydrogenation process, a large amount of heat will be released from the MH bed, causing the bed temperature rises sharply. However, according to the above conclusion, the heat transfer performance inside the reactor becomes the main factor limiting the reaction rate of hydrogen absorption/desorption. Thus, the heat transfer enhancement design of the MH reactors is vital to ensure stable hydrogen absorption and desorption reaction, on which researchers have made lots of efforts. Fig. 3 and Table 1 show the development of some typical MH reactors. Fig. 3a shows paper numbers about the MH reactors in the Web of science database in recent years, it can be seen that it has been on the rise. The author further analyzed the number of thermal management and thermal design in these papers, as indicated in Fig. 3b. It can be found that the paper numbers about both topics are also keep rising in recent years, this result is also confirmed in Table 1. Besides, the paper number about thermal design is higher compared with thermal management, on the one hand, proving the more and more researchers focus on thermal design. On the other hand, the data in Fig. 3b also proves that researchers pay more and more attention to thermal management in recent years. Fig. 3c and Table 1 show some typical MH reactors developed by researchers from all over the world in recent years. It can also be concluded that, in recent years, researchers have not only enhanced the heat transfer of MH reactors by improving thermal design such as optimizing fins, but also enhanced the thermal management capability of the whole system by combining PCM with MH reactors.

In addition, the review papers about the thermal design and thermal management of MH reactors have been summarized and analyzed. Table 2 presents the topic, key points and article information of the corresponding review papers published in recent years. So far, these reviews mainly focus on the topic of enhanced MH reactor thermal design or thermal management respectively. Although the existing literature reviews have been summarized to a large extent, there is a lack of reviews that combine thermal design and thermal management and analyze the future trends of both. As can be seen, This review takes combination of MH reactors and various algorithms as the starting point, and the current situation of MH reactors is summarized and the future development trend is analyzed in terms of thermal design and thermal management, which has remarkable innovation compared with the previous published reviews. Only by combining the thermal design and thermal management optimization of the same MH reactor can the system be more efficient.

3. Thermal design in MH reactor for enhanced performance

3.1. Adding high thermal conductivity materials (HTCM)

3.1.1. Current status

The effective thermal conductivity of the MH particle bed is low (usually 0.1–1 W (m K)$^{-1}$), resulting in high resistance of heat transfer in the bed during hydrogen absorption/desorption, which makes it difficult for the bed reaction heat to exchange with the heat exchange channel in time, and the accumulation of reaction thermal effect will

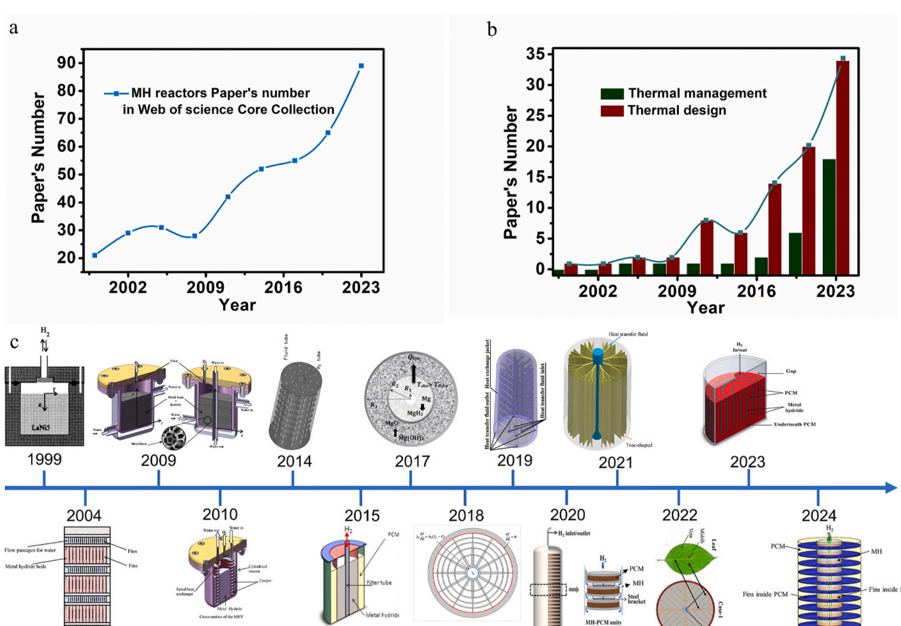

Fig. 3. (a) MH reactors paper number in Web of science Core Collection in recent years; (b) The number of research on thermal management and thermal design in the papers shown in (a); (c) Timeline of typical design cases in MH reactors [44–57].

Table 1
Some typical MH reactors in recent years.

Central theme	Type	Strategy	Alloy	Reactor shape	Year	Ref.	Country
Experimental and theoretical of MH reactor.	Thermal design	Theoretical model	$LaNi_5$	Cylindrical	1999	Jemni A et al. [44]	Tunisia
Heat transfer characteristics of MH reactor	Thermal design	Fin	$Ti_{0.42}Zr0_{.58}Cr_{0.78}Fe_{0.57}Ni_{0.2}Mn_{0.39}Cu_{0.03}$	Cylindrical	2004	Oi T et al. [45]	Japan
Hydrogen storage in MH tanks	Thermal design	Metal foam	$LaNi_5$	Cylindrical	2009	Mellouli S et al. [46]	Tunisia
Experimental study of MH vessel.	Thermal design	Fin	$LaNi_5$	Cylindrical	2010	Dhaou H et al. [47]	Tunisia
Optimization and analysis of heat & mass transfer on MH tank.	Thermal design	Finned multi-tubular	$LaNi_5$	Cylindrical	2014	Ma J et al. [48]	China
Numerical analysis of MH tank.	Thermal management	PCM	MgH_2	Cylindrical and spherical	2015	Mellouli S et al. [49]	Tunisia
H_2 absorption in an MH reactor based on thermo-chemical heat storage.	Thermal management	$MgH_2/Mg(OH)_2$	MgH_2	Cylindrical	2017	Bhouri M et al. [50]	Canada
Heat transfer enhancement in MH reactor.	Thermal design	Fin	$LaNi_5$	Cylindrical	2018	Zhang SW et al. [51]	China
Design optimization and sensitivity analysis of MH reactor.	Thermal design	Mini-channel	$LaNi_5$	Cylindrical	2019	Wang D et al. [52]	China
Numerical simulation on the storage performance of MH tank.	Thermal management	PCM	MgH_2	Cylindrical	2020	Ye Y et al. [53]	China
Enhanced absorption performance of MH device.	Thermal design	Fin	$LaNi_5$	Cylindrical	2021	Bai X–S et al. [54]	China
Bio-inspired leaf-vein type fins for performance enhancement of MH.	Thermal design	Fin	$LaNi_5$	Cylindrical	2022	Krishna KV et al. [55]	India
Performance evaluation of a novel concentric MH reactor.	Thermal management	PCM	$LaNi_5$	Cylindrical	2023	Hassan IA et al. [56]	Egypt
Design and optimization of MH reactor.	Thermal management	PCM	MgH_2	Cylindrical	2024	Shrivastav AP et al. [57]	India

Table 2
Summary of main idea of previous reviews.

Key points of reviews	Field	Country	Time	Ref.	Journal
Optimization and design of different storage bed	MH bed	Russia	2023	Kudiiarov et al. [58]	Materials
Thermal augmentation methods in hydride beds	MH bed	India	2022	Sreeraj et al. [59]	Journal of energy storage
Thermal management techniques in MH	MH reactor	USA	2023	Kukkapalli et al. [60]	Energies
Thermal management technology for MH bed	MH bed	China	2023	Miao et al. [61]	Sustainable energy & fuels
A design methodology of large-scale MH reactor	MH reactor	China	2022	Dong et al. [62]	Journal of energy storage
Development of heat exchanger configuration design in MH bed	MH bed	China	2021	Cui et al. [63]	International Journal of Hydrogen energy
MH thermal management for use in FC system	MH/FC	Australia	2021	Nguyen et al. [64]	International Journal of Hydrogen energy
Design aspects and developmental status of MH	MH reactor	India	2018	Muthukumar et al. [65]	International Journal of Hydrogen energy
Enhanced reaction thermodynamics for hydrogen storage applications	Thermodynamics in MH	South Korea	2018	Kim et al. [66]	International Journal of energy Research
Heat transfer techniques in MH	MH reactor	India	2017	Afzal et al. [67]	International Journal of energy Research
Different reactor and heat exchanger configurations for MH	MH reactor	USA	2016	Shafiee et al. [42]	International Journal of energy
3D modeling and simulation of hydrogen absorption in MH	MH vessels	South Korea	2012	Nam et al. [68]	Applied energy

inhibit the kinetic performance of the hydrogen absorption/desorption reaction and block the process of the hydrogen absorption/dehydrogenation reaction. Adding HTCM into the MH bed can significantly improve the heat transfer performance of MH reactors by increasing the thermal conductivity of bed matrix. The increase of bed thermal conductivity cannot only improve the internal heat transfer efficiency of MH bed, but also promote the heat exchange between the MH bed and the outside environment. The combination of MH alloy with HTCM, such as carbon [69], expanded graphite (EG) [70], metal foam [71,72], copper wire [73], and et al. can improve the effective bed thermal conductivity and promote the hydrogen absorption/desorption reaction [72,74,75]. Wherein, metal foam has high porosity (about 90%), can be used as the bed skeleton, and can absorb part of the stress caused by the expansion/contraction of the alloy during hydrogen absorption/desorption process [76]. At the same time, metal foam has the advantages of high thermal conductivity and large specific surface area. Therefore, adding metal foam to the MH bed can significantly improve the bed thermal conductivity, which has been widely used by researchers [77].

Laurencelle et al. [78] studied the hydrogen absorption/desorption performance of a cylindrical MH reactor embedding metal foam numerically and experimentally. The results showed that the effective thermal conductivity of the MH bed increased from 0.15 W $(m\ K)^{-1}$ to 10 W $(m\ K)^{-1}$ after the addition of aluminum metal foam to the reactor. Similarly, Ferekh et al. [72] studied that the hydrogen absorption rate of the metal foam reactor is better than that of the fin reactor because the thermal conductivity of the bed is increased and the temperature uniformity of metal foam MH bed is better. Mellouli et al. [46] studied the effect of embedding metal foam into MH bed on the hydrogen absorption performance of different MH reactors. The results indicate that the time required for the bed hydrogen concentration to reach 90% for the

MH reactor is reduced by about 60% after adding metal foam, and the improved hydrogen absorption rate for the central heat exchange tube reactor by adding metal foam is significantly better compared with the water jacket reactor. The above results show that the MH reactor has better performance due to the better temperature uniformity of the bed after adding metal foam. Moreover, some scholars further studied the effect of metal foam filling ratio on the performance of hydride reactors. For example, Wang et al. [79] numerically studied the evolution of hydrogen absorption properties of metal foam reactors with the volume fraction of aluminum metal foam under different cooling conditions. The results show that under the active cooling condition, only 10% volume fraction of metal foam can play a high heat transfer performance. Tsai et al. [80] researched the evolution of hydrogen absorption performance of an MH reactor with the volume fraction of aluminum metal foam while keeping the filled MH alloy in the bed constant. They concluded that adding only a 1% volume fraction of aluminum foam to the MH bed can significantly improve the hydrogen absorption rate.

Above all, it is illustrated by examples that the embedded metal foam uniformly improves the effective thermal conductivity of MH bed. It not only enhances the heat discharge from MH bed to heat transfer fluid, but also improves the uniformity of heat transfer in MH bed. Therefore, it is necessary to study the optimal arrangement of high thermal conductivity materials embedded in the bed to improve the heat transfer performance of the reactor bed while reducing the introduction of additional heat transfer system mass and volume. There is little research has been reported on this.

Some scholars have combined MH alloys with other high thermal conductivity materials to improve the thermal conductivity of the bed matrix. Kim et al. [5] prepared MH composites by copper encapsulation and cold pressing of MH particles ($LaNi_5$) in a low-melting metal binder. The measurement results showed that the thermal conductivity of MH bed is increased by 5 W $(m\ K)^{-1}$. Dieterich et al. [81] formed composite press blocks with MH alloy and expanded graphite, and experimentally studied the hydrogen absorption/desorption properties of the MH reactor packed with composite press blocks. It was indicated that the thermal conductivity of the composite block can reach 40 W $(m\ K)^{-1}$, but it will decrease to 10–15 W $(m\ K)^{-1}$ after 1000 cycles of hydrogen absorption and desorption. Zhu et al. [82] synthesized MgH_2-EGN composite compact, and the results found that the absorption/desorption hydrogen rate of the hydride reactor increased significantly as the proportion of expanded graphite increased. Generally, the porosity of metal foam in the research is evenly distributed and completely filled within the bed. It should be noted that the embedded high thermal conductivity material will occupy a part of the available volume of the MH bed, resulting in a decrease in the hydrogen storage capacity of the MH reactor [83]. Therefore, it's great significance to optimize the arrangement of high thermal conductivity materials in MH reactors in a certain quantity. In this way, the amount of highly conductive materials in the MH bed can be reduced, while the heat transfer strengthening ability remains unchanged. Bai et al. [84] proposed a fin-metal foam MH coupling reactor (FMF reactor). The FMF reactor was studied by using a 3D multi-physics model. The ratio of metal foam in the FMF reactor was optimized by genetic algorithm (GA) under a certain amount of high thermal conductivity material. The result showed that the hydrogen absorption time of the optimized FMF reactor is 6.9% and 38% shorter than that of the MF reactor and the Fin reactor, respectively. The results show that high thermal conductivity materials, the optimal fin volume ratio of the optimized fin-metal foam reactor is maintained at about 0.4, under different volume fractions.

Through the above summary, it is found that many studies still choose the best value within a certain range of data, and then to conclude. However, this exhaustive method is not a means to solve the problem. Combined with the latest research, machine learning (ML) is combined with traditional research methods in the limited data to obtain an optimal value, so that the obtained value is accurate and avoids a large number of cumbersome calculations, which is also a trend for future research.

3.2. Design and optimization of heat transfer fins

Embedding fins into the MH bed is another effective method to improve the hydrogen absorption/desorption rate of MH reactors. On the one hand, fins can increase the heat transfer area and decrease the heat transfer distance of the MH alloy; on the other hand, fins, as efficient heat conduction channels, can effectively promote the heat exchange between the MH bed and the heat exchange channel, thereby increasing the absorption/desorption hydrogen rate [85].

The embedding fins can be divided into internal fins and external fins. The internal fins are used to improve the heat transfer performance of the MH bed, while the external fins are generally applied to enhance the heat exchange between the reactor and outside environment when the natural convection boundary is adopted.

3.2.1. Internal fins

The internal fin can not only increase the contact area between the hydrogen storage material and the cooling pipe, but also improve the local thermal conductivity of the hydrogen storage bed. The inner fins can be divided into transverse fins and longitudinal fins in form [86]. From the data of simulation and optimization results, the longitudinal fin optimization design seems to be more effective than the transverse fin optimization design, but the efficiency gap is negligible in the actual situation. Therefore, it can be considered that the two fin designs have the same effect on improving heat transfer performance.

Moreover, the parameters of the fins were studied by the researcher Ma et al. [48], as shown in Fig. 4a. They found that the thickness of the fin has a direct effect on the heat transfer efficiency. When the thickness of the fin increases, the thermal resistance decreases, the thermal conductivity increases, the heat dissipation effect of the whole device is improved, and the hydrogen absorption rate of the device is greatly increased. They also found that an increase in the number of fins significantly reduced the maximum average temperature. Similarly, improving the fin radius can also improve the heat transfer effect. Similarly, Anurag et al. [87] investigated the performance of a 1 kg MH reactor embedding finned tube heat exchanger. The study looked at the effects of different structures, with absorption times reduced by 570 s when the number of fins increased from 4 to 13. When the diameter was increased from 3.4 mm to 5.4 mm, the reduction was 210 s, while the effect of fin perforation was minimal. The results of the Gkanas [88] team's and Aashish [89] team's research work also show the similar conclusion as above. There have also been studies of longitudinal fins. For example, Bhouri et al. conducted numerical studies on the optimization of longitudinal finned multi-tube MH reactors [90]. The results show that the installation of longitudinal fins can help the reactor save 41% of hydrogen absorption time compared with the situation without longitudinal fins. The conclusion about fin parameters is the same as that of transverse fin which is mentioned above.

The above literature shows that the thickness, number and radius of fins are important parameters that affect the overall heat transfer effect, which suggests that we should comprehensively consider the parameters of fins when designing fins, to achieve the best heat transfer effect. In addition, the material of the fin also plays a vital role in the heat transfer effect, so the researchers studied the material of the fin.

The choice of fin material also has a great impact on the heat transfer performance of the MH reactor. Nyamsi et al. [91] compared the hydrogen absorption performance of the MH reactors embedding different material fins including aluminum, stainless steel and copper. The results show that although the thermal conductivity of copper fins is the highest among three material fins, the thermal conductivity of aluminum fins is better than that of copper. Moreover, copper fins will increase the cost of system and weight of the overall structure. Therefore, they suggest the actual use of aluminum fins which have high thermal conductivity, low density and low cost. This also inspires

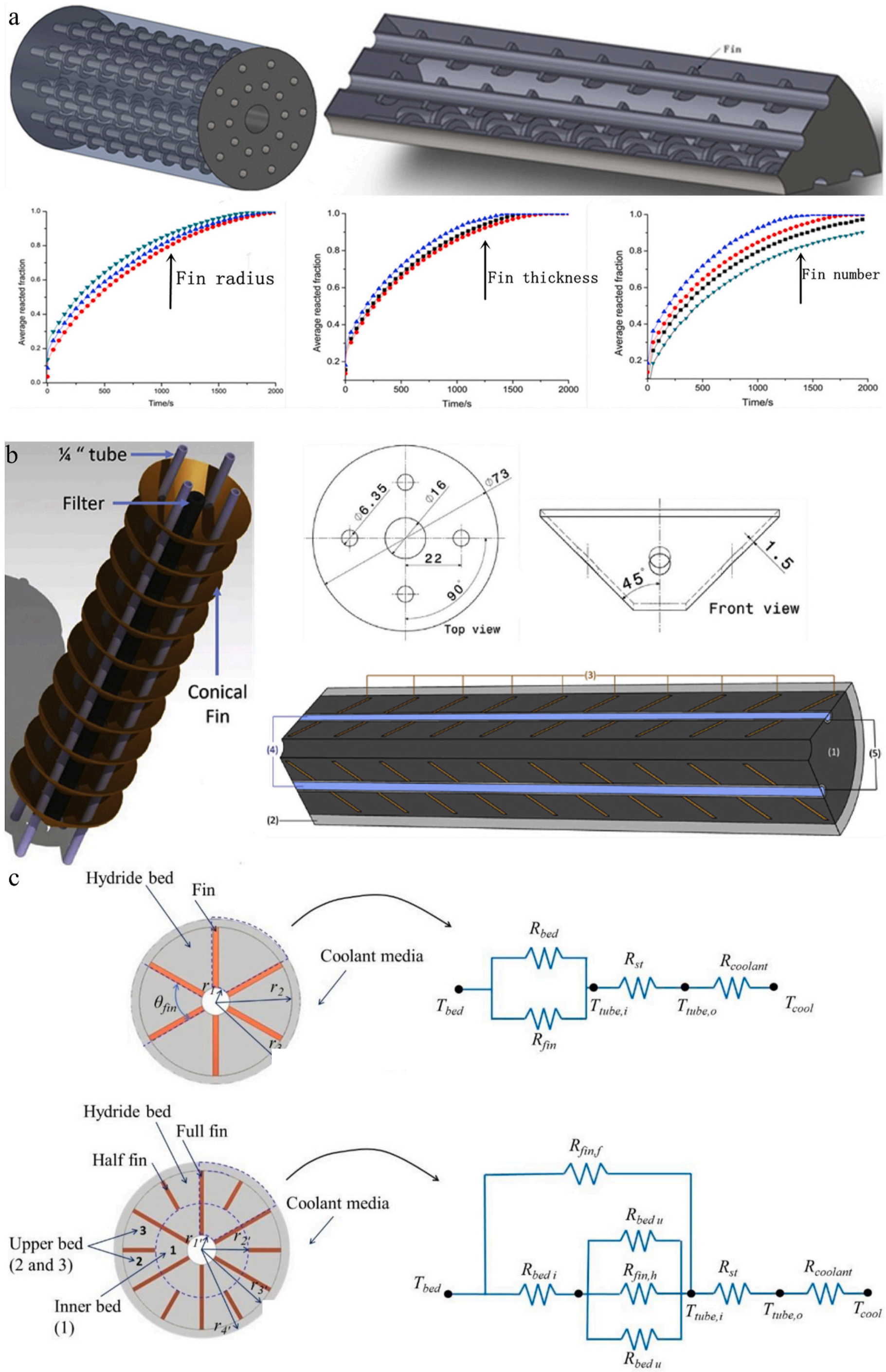

Fig. 4. (a) The schematic of fined multi-tubular MH tanks [48]; (b) The figure of conical copper fins [92]; (c) Cross-section of single MH tube and corresponding thermal resistance network: case with 6 full fins (up) and case with 6 pairs half and full fins (down) [95].

researchers or technicians to apply aluminum fins in practical applications in engineering, because these advantages which mentioned above.

In addition to the traditional fin shapes mentioned above, researchers have proposed reactors with different new fin shapes. For example, Chandra et al. proposed a new type of conical fin reactor [92, 93]. As shown in Fig. 4b, there is a central filter inside the reactor that evenly supplies hydrogen to the reactor. The conical copper fins are passed through four cooling tubes to carry away the heat. The advantage of this new fin is that it is simple to install and can provide a larger surface area to enhance heat transfer. The simulation results show that the reactor with conical fin design has a larger specific surface area than the reactor with round fin design, so the heat and mass transfer performance is better. At the same time, the simulation results also prove that the conical fin and the circular fin have similar trends in the sensitivity analysis of structural parameters such as fin number and fin spacing. Furthermore, in their other published literature, it is also concluded that while increasing the number of fins, the thickness of fins should be correspondingly reduced to improve the performance of MH reactors [94]. This was done to minimize volume and density loss, and the results showed a significant increase in the MH bed's storage rate and a significant reduction in the time required to reach the 90% saturation level. To minimize the impact of fin volume on hydrogen storage while increasing heat dissipation, Pandey et al. [95] used the Thermal Resistance Network analysis method to optimize the fin distribution in a MH bed with a fixed total fin volume, as shown in Fig. 4c. The thermal resistance concept introduced can be used to optimize any traditional fin-based hydrogen storage reactor to obtain the best heat transfer performance. Wang et al. [96] proposed a corrugated finned-tube reactor, the hydrogen absorption time for 80% saturation of the corrugated finned-tube reactor is 32% lower than that of the traditional disc finned-tube reactor.

3.2.2. External fins

In some applications (such as cars and vessels) the available system space is small, and thus the volume of the MH reactor needs to be reduced. Using the natural convection condition rather than the heat exchange tube can decrease the reactor volume. The external fins are generally embedded on the outer wall of the MH reactor to increase the convective heat transfer area. During hydrogen absorption, the reaction heat transfer relies on the internal MH to transmit heat to the tube wall first, and then dissipates heat by free convection. However, it does not achieve good heat transfer effect [97,98], as shown in Fig. 5a. Therefore, Elhamshri et al. [99,100] used $LaNi_5$ to conduct experimental and numerical simulation studies on hydrogen absorption performance of three different cylindrical MH reactor configurations, as shown in Fig. 5b. The high-temperature zone in the middle of the reactor can transfer heat to the top natural convection boundary by heat pipes, rather than heat transfer to the surrounding boundary by heat conduction of MH. Moreover, the use of fins increases the heat dissipation area of natural convection boundary, which has a significant effect on improving the heat and mass transfer process in MH reactor. The experimental data and simulation results show that the absorption rate is increased by about 70% under the same hydrogen supply pressure. Meanwhile, Tetuko et al. have proposed a similar configuration, as shown in Fig. 5c [101]. In their research, they proposed a forced convection boundary by mounting a blower on the side of the fin. Increasing the interaction between the fins and the fan makes the heat loss of the fuel cell faster and also provides convenience for reducing the size of the container. Although the number of studies on the external fins is less than that on the internal fins, it lays a foundation for the study of heat transfer using the fin in small space. David et al. [102] use external stainless steel tubes and external aluminum fins to form MH reactor, which are combined with proton exchange membrane fuel cells (PEMFC) to achieve the purpose of utilizing PEMFC waste heat. The forms of different fins have their own appropriate uses, so it has a certain practical significance. Khayrullina et al. [103] developed a new hydrogen storage reaction using the H2Smart experimental device, which uses the waste heat inside the FC system to carry out hydrogen desorption process instead of the external heater. Tetuko et al.'s [104] innovative idea of thermally coupling PEMFC and MH reactor using heat pipes provides an opportunity to reduce costs and improve performance and compactness. Similarly, Mahmoodi et al. [105] conducted an experimental study on the coupling of fuel cells with MH tanks and investigated the effects of the number of heat pipes, the number of wires, and the types of heat pipes with and without fins. In addition, Parida et al. [106] found that at best only 16–17% of the waste heat of PEMFC is utilized to desorb a single MH reactor, founding theoretical boundary values. Gao et al. [107] proposed an innovative PEMFC-MH coupled cold start method, which shortened the cold start time (101 s) by 59.3% and reduced the cold start energy consumption by 62.4% compared with the traditional method, and achieved significant progress in saving time, energy consumption and hydrogen utilization. Moreover, the coupling system of solid-state hydrogen storage and fuel cell developed by Yao et al. [108] effectively improves the utilization efficiency of hydrogen storage tank and fuel cell. The results of the road test study show that the hydrogen-assisted two-wheeler can travel 80 km at a speed of 25 km per hour. These actual test data provide more powerful support for the practical operation of hydrogen fuel cell vehicles.

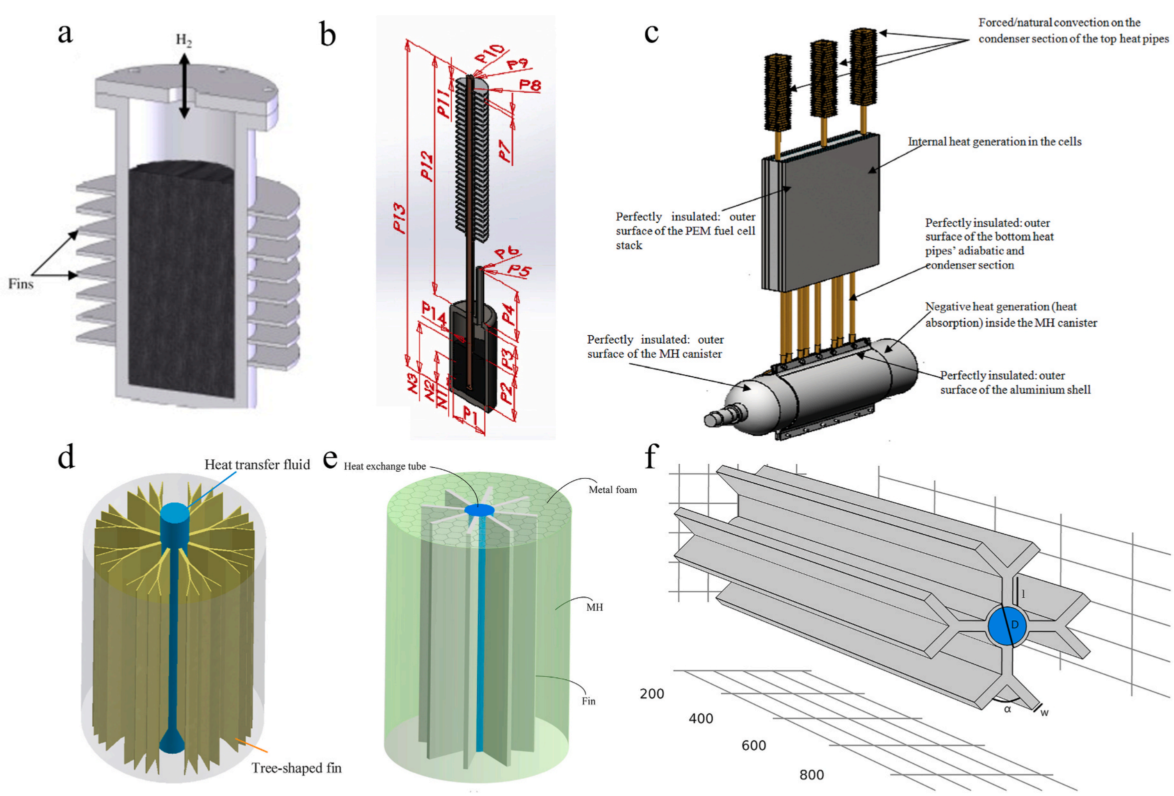

Fig. 5. (a) Researched by Askri et al. [98]; (b) Geometries of the MHRs with finned heat pipe used in modeling and simulation [99]; (c) The numerical model of a thermal coupling [101]; (d) The schematics of the tree-shaped fins reactor [54]; (e) Schematics of the coupled fin-metal foam MH reactor (FMF reactor) [84]; (f) Heat exchanger with longitudinal bifurcating fins [89].

3.2.3. Development tendency of design and optimize fins

From above, the fins have been widely applied for improving the heat transfer inside MH bed or enhancing the natural convection heat transfer by connecting with the outer wall of MH reactors. However, the current researches on fin design have the following shortcomings: On the one hand, fin structure design is mostly based on traditional heat transfer theory and lacks the introduction of novel design concepts, resulting in a lack of significant innovation in fin structure and limited improvement in the hydrogen absorption/desorption performance of the MH reactor. On the other hand, the current research on fin structure and layout parameter optimization mainly adopts the enumeration method, which is low optimization efficiency and easy to be influenced by the parameter setting range, so it is difficult to accurately find the optimal value. The traditional research method has been very limited to the point that it must be improved. For example, in Fig. 5d, Bai et al. [54] proposed a tree-shaped fin to improve the heat transfer area of the side fin. In this review, based on bionic analysis, vein fins are used to enhance heat transfer in hydride bed. The dynamic hydrogen absorption process in the hydride reactor was simulated numerically, and the mechanism of enhancing the bed heat transfer efficiency was revealed. Furthermore, the structure of vein fins was optimized by GA with the minimum fire deposition dissipation thermal resistance as the objective function. The optimal configuration evolution of vein fins under different length coefficients, maximum bifurcation level and bed thermal conductivity and the hydrogen absorption rate of the corresponding hydride reactor were studied. Numerical simulation results indicate that the response speed of the optimized model is improved by about 20% compared with the traditional longitudinal fin. It is a remarkable fact that they also proposed the high-thermal material coupling fin-metal foam arrangement scheme by the same method combined with GA, Fig. 5e [84]. The thermal resistance of the coupled fin-metal foam bed, finned bed and metal foam bed was calculated theoretically, and the effects of the three material arrangements on the hydrogen absorption performance of the hydride reactor were compared. Then, GA was used to optimize the material distribution and corresponding structural parameters under different volume proportions of highly thermally conductive materials, fin number and fin structure, and the evolution of the influence of volume proportion of highly thermally conductive materials on the hydrogen uptake rate and hydrogen storage capacity of the reactor was analyzed in detail. When VFHM was 0.08, the comprehensive hydrogen uptake performance of FMFR was optimal [84]. Aashish et al. [89] proposed a cylindrical reactor design with an embedded bifurcated longitudinal fin cooling tube as a heat exchanger for an industrial-scale MH reactor system, as shown in Fig. 5f. The design parameters were optimized by Taguchi method, and the performance of the model was simulated with the minimum absorption time as the objective. After optimization, the optimal parameters of the number of cooling tubes, the length of fins and the number of fins were obtained.

The above cases show that using the algorithm to calculate the parameters of the fin can get the best value so as to achieve the best heat transfer effect. However, from the current point of view, there is no great breakthrough in the shape design of the fin, so the author suggest that the method of topology optimization can be adopted to optimize the fin, so as to achieve the best fin shape in the system. For example, He et al. [109] adopted a topology optimization strategy to optimize the spatial layout of fins in order to maximize the thermal performance of LHTES units. Similarly, wang et al. [110] proposes an enhanced topology optimization model to maximize the heat transfer performance of fin structures embedded in PCM. All the above studies show that topologically optimized fins can exert the best heat transfer performance. At present, topology optimization strategy has not been widely applied in MH reactor, and the author believes that topology optimization will be innovatively applied in MH reactor for a long time in the future. For instance, Shi et al. [111] demonstrate that topological optimization effectively overcomes the limitation of low thermal conductivity in $MgO/Mg(OH)_2$. In comparison to a non-finned reactor, a triangular fin reactor, and a longitudinal fin reactor, the topologically optimized structure exhibits remarkable heat transfer capability. Notably, the topological structure incorporating copper-containing materials significantly reduces hydrogen absorption time. Krishna et al. [112] optimized the fins to achieve the best heat transfer performance and improve the MH reactor reaction rate. The aforementioned statement undoubtedly signifies a positive indication, signifying the initial achievements in employing the topological optimization method in MH reactors and establishing a solid groundwork for future researchers to further delve into this field.

3.3. Heat exchange tube design in MH reactors

3.3.1. Current status

In the process of hydrogen absorption, the reaction heat in MH bed needs to be dissipated to the outside in time, while it is necessary to input enough heat from the outside to the bed to ensure the efficient dehydrogenation reaction during hydrogen desorption. Generally, the natural convection or forced convection heat transfer condition are adopted to achieve heat exchange between the hydride bed and the outside environment, and the heat transfer media are air and water (the heat transfer medium is oil in high temperature MH reactors), respectively. In Fig. 6a, Sekhar et al. [63,113] studied the heat transfer properties of cylindrical MH bed with various heat transfer schemes. The hydrogen absorption rate for different cases can be ranked as: Vertical tube < Cooling jacket/no fin < Spiral tube ≈ Cooling jacket/transverse fins. The inner spiral tube reactor has better absorption performance than the inner straight tube reactor and the outer cooling jacket reactor. However, when sufficient copper fins are added to the MH reactors, the same improvement can be achieved in the external cold-jacket model. Wu et al. [114] made a similar comparison between the straight tube and spiral tube in magnesium MH reactors. Chang et al. [115] used spiral fins to enhance the heat transfer in the bed of the water-jacketed hydride reactor. The numerical results show that the hydrogen absorption rate of the reactor with spiral fins is significantly higher than that with longitudinal fins when the volume of fins is the same. The simulation results showed that the spiral tube can improve the heat transfer more significantly than the straight tube because the larger heat transfer area of spiral tube and the secondary flow. At the same time, for hydrogen desorption process, spiral tube is still a better choice. For example, Wang et al. [116] used a double helix structure to transfer PEMFC waste heat to a metal hydride tank, obtaining a coupled system with an efficiency of up to 81.7% and a stable working time of 6790 s. Because the spiral tube not only increases the heat transfer area, but also greatly increases the heat transfer coefficient due to the secondary flow in the spiral tube, and finally achieves better heat transfer effect than the straight tube, spiral tube has received extensive attention by researchers. Moreover, a cylindrical storage tank with coil heat exchanger structure can also be developed by combining the optimization ideas of other reactors. The spiral tube has been proven to effectively enhance the heat exchange between the MH bed and the heat exchange fluid in MH reactors. Eisapour et al. [117] inserted a center return tube into a primary spiral heat exchanger. The new structure has a better absorption rate than the original structure, and the absorption time is reduced by 24%.

Besides the decrease of pitch and the increase of spiral coil diameter can increase the heat transfer area. Semi-cylindrical coil heat exchanger (SCHE) is developed from the helical coil heat exchanger (HCHE) and is located in the MH reactors [120]. The study demonstrated that using this heat exchanger can significantly reduce hydrogen absorption time by 59% compared to the HCHE type. However, the MH bed center is far away from the HTF region and the temperature is higher, resulting in a lower hydrogen concentration in this region. Therefore, in order to improve the heat exchange rate in MH reactor central region, Larpruenrudee et al. [121] developed a SCHE with center return tube (SCHE-CR) on the basis of a SCHE. Numerical simulation results show

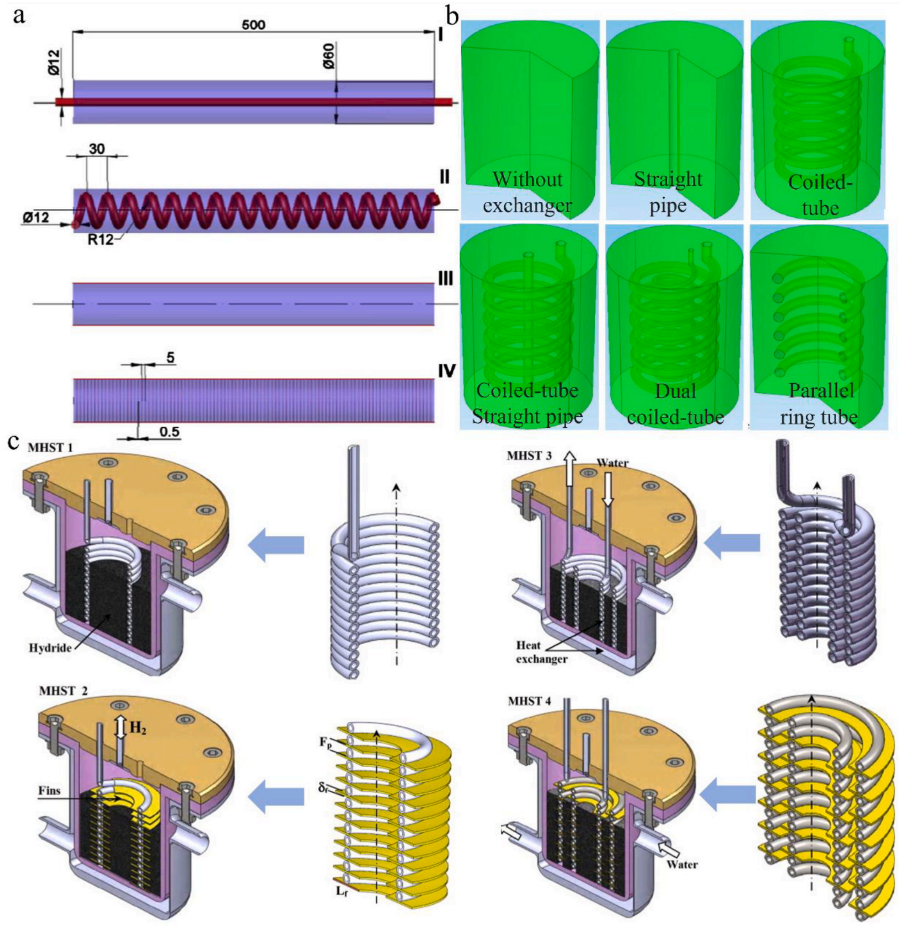

Fig. 6. (a) Schematic drawings, average hydrogen concentration and temperature of MH bed with various heat exchange options [63]; (b) Diagram of different MH reactors [118]; (c) Schematic of MHSTs used in simulations [119].

that the absorption time of SCHE-CR is reduced by 30% compared with SCHE. By increasing the diameter of SCHE-CR, the absorption reaction rate was increased by 40%. The optimal structure proposed by Feng et al. [122] is U-shaped double helix tube. Compared with the conventional spiral tube, the U-shaped structure can increase the heat transfer area. Unlike the double helix structure, the U-shaped has only one inlet/outlet for easy installation in the MH bed. In order to effectively remove heat from the central area, Liang et al. [118] simulated different MH reactors filled with $LaNi_5$ by combining cooling tubes of different shapes, as shown in Fig. 6b. The final results showed that it took only 271 s for dual coiled-tube to achieve a 90% hydrogen absorption rate, while others needed more time to reach the target. Dhaou et al. [47] designed a finned spiral heat exchanger by combining the straight tube and spiral tube. The structure and operating parameters of the heat exchanger were experimentally studied to determine its influence on heat transfer performance. Mellouli et al. [119] proposed the double-fin spiral tube MH reactor, as shown in Fig. 6c. The results show that MHST4 is much faster and the utilization of circular fins can effectively improve the absorption rate of the reactor by nearly 60%. We can come to a common conclusion that, increasing the heat transfer surface area and reducing the heat conduction distance of the MH alloy, can increase the heat transfer rate and making the absorption performance better.

Mohan et al. [123] theoretically analyzed the use of heat exchanger tubes in MH bed and found that pipe diameter had little effect on hydrogenation time. Muthukumar et al. [124] proposed a multi-tube heat exchanger in an MH reactor, investigated the effect of the number of cooling tubes on the hydrogenation time, and found that 80% of the total absorption capacity was completed in 300 s with the use of 12 cooling tubes. But in contrast, when 20 tubes were used inside the bed, the same absorption took just 100 s. They also found that increasing the number of tubes to more than 20 did not affect absorption time. Similarly, Anbarasu et al. [125] also found through numerical calculation that increasing the number of cooling tubes to a certain limit can reduce the absorption time, but after increasing to a certain limit, it has no significant impact on the system performance. Mahmoodi et al. [105] also conducted a similar study on the coupling of fuel cell and hydrogen storage tank, and reached a consistent conclusion. Some scholars have also conducted research on a large number of straight tube MH reactors. For example, Jana et al. [126] designed a shell and tube MH reactor composed of 99 tubes and 10 mm outer jacket to integrate with LT-PEMFC with a capacity ranging from 1 to 3.5 kW. Li et al. [127] proposed a novel spiral micro-channel reactor to improve the heat transfer efficiency of MH bed. Moreover, the structural parameters of the spiral channel are also studied. It is concluded that under the fixed reaction fraction value, the larger the radius the better the heat transfer effect, and the smaller the axial pitch the faster the heat transfer rate. This is mainly due to the increase of the radius, the increase of the heat transfer area, and the decrease of the axial spacing, the decrease of the cooling boundary distance.

To further expand the advantages of multi-tube bundle reactors and maximize the heat transfer effect, some researchers have further done research based on multi-tube bundle reactors. For example, Wang et al. [52] proposed a new type of radiation mini-channel reactor (RMCR) to improve thermal efficiency, and developed a RMCR with jacket to eliminate the adverse heat transfer zone inside the RMCR. The simulation results show that when the pressure is 1 MPa and the initial fluid temperature is 293 K, the reaction time of radiant tube is 77%, 52% and 37% less than that of straight tube and spiral tube respectively but the performance is the best. Based on this radiation tunnel, their research team proposed a new structure by adding branch channels, called branch mini-channel reactors [128]. Because this new structure, the heat transfer effect is further enhanced, which can complete the hydrogenation/dehydrogenation process faster. Overall, at present, the multi-tube reactor has made great progress in enhancing the heat transfer effect. The radiation tube heat exchanger can significantly improve the hydrogen storage performance of MH bed, and may be the best choice for the current MH reactor cooling tube design.

3.3.2. *Development tendency*

The structure design of heat exchange tubes is necessary for enhancing heat transfer and improving the hydrogen storage performance of MH reactors. Scholars have used many structures and methods to improve the heat transfer rate inside the MH reactor, among which there are many studies using spiral tubes. Therefore, it is necessary to conduct experimental research on the MH reactor with embedded spiral tubes.

As shown in Fig. 7, Saurabh et al. [129] made an MH reactor using $LaNi_5$ as a metal hydride and water as a heat transfer fluid flowing through embedded spiral tube and conducted experimental research. The results show that the hydrogen absorption rate can be accelerated with the increase of hydrogen supply pressure, mass flow rate and heat transfer fluid temperature. Similarly, an increase in mass flow and an increase in the temperature of the heat transfer fluid will also lead to a faster rate of hydrogen desorption. Based on the experimental data, various ML models were developed and tested, and these models were then used for performance prediction. The study found that the tree-based regression ML model is superior to other models. Therefore, the model can be used to predict hydrogen uptake or desorption rates and MH bed temperatures at different parameter values without the need for detailed experiments and numerical tests. Finally, the system is optimized by brute force algorithm for three different applications. Optimization techniques help determine parameter values for specific applications, saving multiple experimental runs of adsorption and desorption. This is a new effort to model and optimize MH reactors using ML, and using these tools can save a lot of computational time, effort and experimental resources. Kanti et al. [130] used the evolutionary ML technique of gene expression programming (GEP), building a prediction metamodel based on data acquired in the numerical investigations. Suarez et al. [131] used Differential Evolution (DE), Particle Swarm Optimization (PSO), and Genetic Algorithm (GA) which used to determine the parameters of a commercial metal hydride hydrogen storage tank to achieve an energy management strategy.

In addition, not only MH reactor's structural design, ML is already being used in many other fields [132]. ML has also been used to coupling MH to fuel cell systems by Zhou et al. [135], greatly reducing costs during research. Tian et al. [133] used artificial neural network (ANN) for cycle life prediction to determine the durability of magnesium based hydrogen storage alloys due to substandard durability. Nations et al. [134] used ML for hydrogen storage and material design. Suwarno et al. [136] used ML to reveal the effect of alloy composition on the hydrogen absorption characteristics of AB_2 alloy.

The above research provides a new direction for us to combine ML with traditional simulation and experimental research. Due to the heavy calculation cost of traditional CFD model, the velocity of solving the complex multi-field coupling process between multiple components is very low, which makes the efficiency of studying the thermal management and thermal design of MH reactor very low. However, ML saves a lot of computing time, energy and experimental resources. And also provides help for us to continue follow-up research. We foresee that the partnership between MH reactors and ML will lead to the design of novel and modified systems.

4. Thermal management of MH reactors

4.1. *Using phase change materials*

During hydrogen desorption, a large amount of reaction heat needs to be input into the MH bed to maintain the dehydrogenation reaction, which will increase the energy consumed for hydrogen storage system. Due to that the hydrogen absorption process in MH bed will release lots of reaction heat, it is necessary to accumulate the hydrogenation reaction heat and use it for hydrogen desorption process to improve the system energy efficiency. Phase change materials (PCM) are considered to have good thermal management ability because of their high thermal

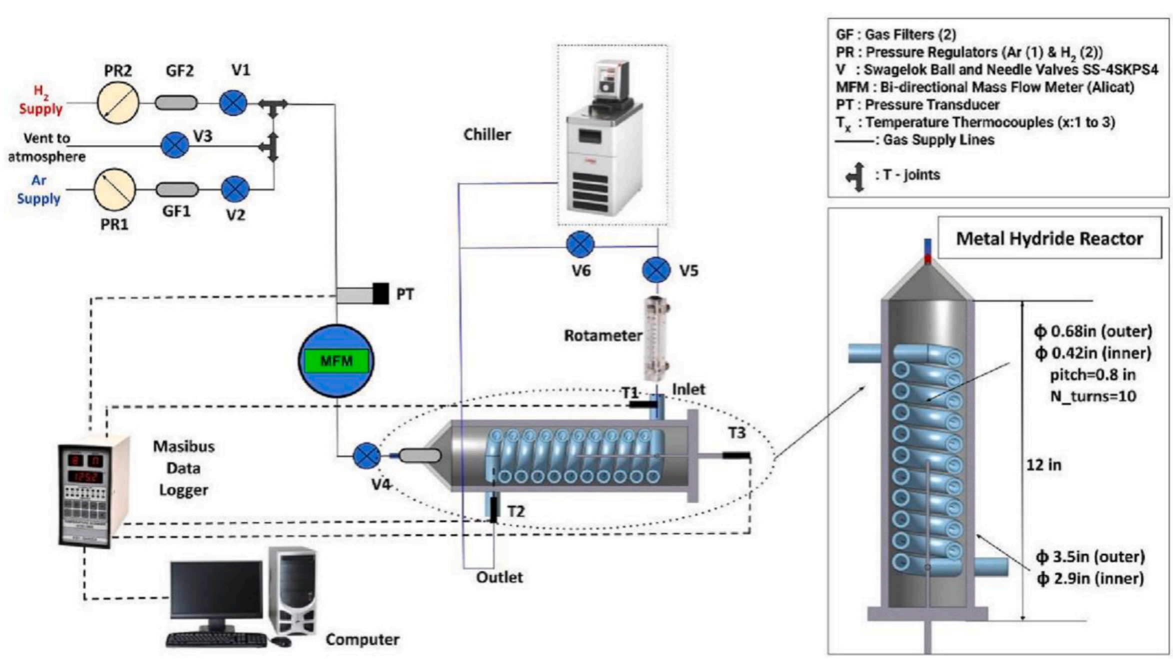

Fig. 7. Line diagram of experimental setup and detailed specifications of MH reactor [129].

storage capacity and isothermal properties [137]. Therefore, are trying to solve this problem by using PCM in hydrogen storage systems [138]. That is to say, the heat released during the absorption process is transferred to the PCM, which can be reused in the subsequent dehydrogenation process [139–141]. For example, Nyamsi et al. [142] demonstrated that the heat storage capacity of PCM can be used as a range extender in the coupling system of a metal hydride reactor and fuel cell in light-duty fuel cell vehicles. Yao et al. [143] used PCM to improve the hydrogen storage efficiency and made the MH reactor FC coupling system perform well. After 10 cycles, the performance of the power system did not decrease significantly.

The layout scheme of PCM in the MH reactor is of great influence on its thermal management performance, and some schemes have been proposed. Mellouli et al. [49] constructed a cylindrical MH reactor surrounded by a PCM jacket, and numerically simulated the effect of reactor parameters, in Fig. 8a. The results showed that, with this arrangement, 80% of the total stored hydrogen can be released without the use of any external heat source, which confirmed the feasibility of

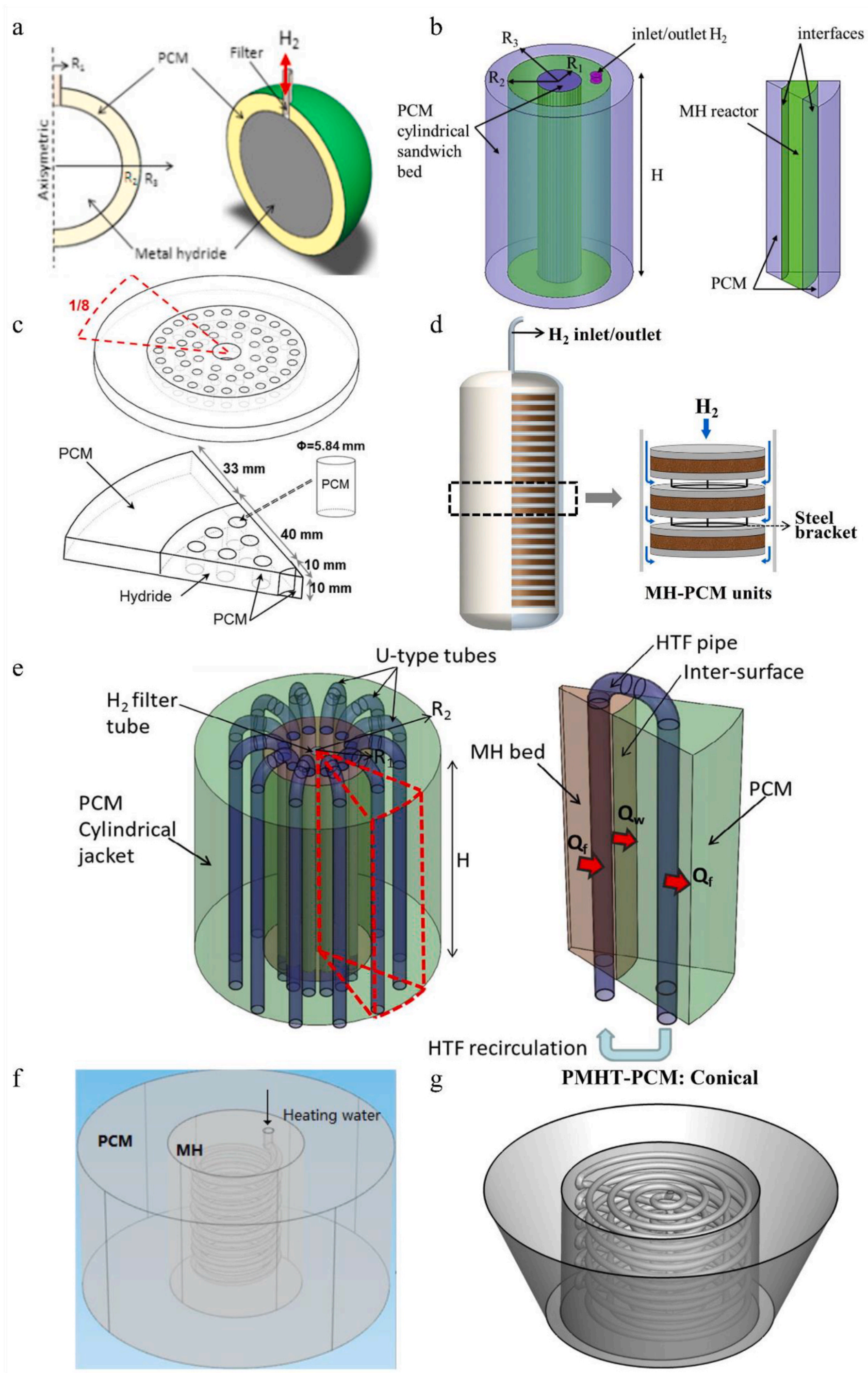

Fig. 8. (a) Optimization of MH bed designs with PCM: spherical tank [49]; (b)MH encircled by a cylindrical sandwich bed packed with PCM [144]; (c) The MH disc is stacked coaxially on a cylindrical tube filled with PCM which arranged concentrically [145]; (d) The cylindrical hydrogen storage tank with MH-PCM units [146]; (e) Geometries of various configurations of MH-PCM tank with closed HTF pipe [150]; (f) Diagram of MH tank with hybrid heat transfer consisted of PCM jacket and coiled-tube [151]; (g) MH reactor of conical [152].

PCM thermal management for MH reactors. Mghari et al. [140] found that the thermal conductivity of PCM was positively correlated with the rate of reversible hydrogen desorption process. The higher the thermal conductivity, the lower the reaction time. It should be noted that the heat storage capacity (melting enthalpy) of PCM has a greater influence on the reaction time. Many researchers optimized the combined structure of MH reactors and PCM to increase the heat transfer area, reduce the thermal resistance and improve the reaction rate [49]. To increase the heat transfer area, sandwiched MH reactors between PCM (both inside and outside PCM, as shown in Fig. 8b) [144]. The reaction time of MH bed with PCM was significantly reduced by 81% and 74%, respectively. To provide better heat transfer conditions, Mellouli et al. [145] inserted a small encapsulated PCM tube in the middle of the MH bed while keeping the PCM layer around the MH bed, as shown in Fig. 8c. According to the results, this method helps to improve the hydrogen charging time by more than 58% compared to the most common arrangement. However, this approach is expected to reduce the hydrogen storage capacity of MH devices. In addition, the encapsulated PCM tubes showed signs of damage due to the expansion and contraction of the MH bed. In order to solve this problem, Ye et al. [146] proposed another sandwich structure, as shown in Fig. 8d. The whole system was stacked by MH-PCM units (PCM sandwiched between two layers of MH), and a hydrogen tunnel was added in the center of the system. Compared to the external jacket, the sandwich structure can provide a higher thermal conductivity surface, resulting in a faster hydrogen adsorption/desorption rate.

According to the literature analysis, it is evident that the combination of PCM and MH reactor yields favorable results in terms of accelerating the reaction rate. However, it is crucial to note that the design structure should not compromise hydrogen storage space at the expense of reducing its capacity, as this trade-off would be counterproductive. Instead, a more effective approach would involve minimizing each hydrogen storage unit's size to prevent any adverse impact on both reaction rate and hydrogen storage capacity.

Due to the low thermal conductivity of PCM, the heat transfer in PCM also needs to be improved. Adding high thermal conductivity materials in the PCM is an efficient method. Darzi et al. [147] studied the effects of operating parameters such as the porosity of MH bed, hydrogen pressure and thermal conductivity of PCM on the heat transfer characteristics and reaction performance of MH reactor with PCM. The results show that the addition of metal foam can improve the thermal conductivity of PCM jacket and increase the melting and solidification speed, resulting in the shortening of adsorption and desorption time of MH tank. Similarly, Tong et al. [148] attempted to insert aluminum and copper foams into PCM to increase thermal conductivity. Because the thermal conductivity of copper foam is higher than that of aluminum, the PCM composed of copper foam has better heat and mass transfer performance, resulting in less reaction time for the MH reactor. And, the addition of expanded graphite to the PCM also significantly improved the performance of the MH reactor coupled FC system, as demonstrated by the work of Yao et al. [149].

Optimizing the design of heat-exchange tube is another feasible method to enhance the heat exchange between the MH bed and the PCM. Mellouli et al. [150] applied the U-shaped tube to enhance the heat transfer between the MH bed and PCM, as shown in Fig. 8e. The U-shaped tube through both the PCM and the MH bed, and forms a flow loop. The cooling medium flows through these U-shaped tubes, removing the generated heat from the MH bed and transferring it to the PCM. In this case, due to the help of the U-shaped cooling tube, the heat transfer rate to the PCM is increased, and the heat loss is reduced, thus improving the hydrogen and heat storage performance of the system. It has been proved that the heat exchange efficiency of the spiral tube is excellent. Therefore, such tubes can be used to improve the performance of the reactor with PCM. Tong et al. [151] proposed a new type of heat exchanger based on the combination of PCM jacket and spiral tube, as shown in Fig. 8f. The hydrogen desorption process is studied, and the results show that a faster desorption rate can be obtained due to the. Unlike the previous U-shaped design, the heat transfer process of the PCM jacket and the spiral tube is independent of each other. This limitation causes some heat loss during hydrogen absorption.

In order to enhance the uniformity of heat transfer between the top and bottom, the PCM layer in the outer coating is specifically designed in a conical shape to optimize heat transfer from top to bottom. Ardahaie et al. [152] proposed a conical PCM jacket structure, Fig. 8g. The performance of spiral tube conical PCM jacket in the process of adsorption and desorption is better than that of cylinder model. The geometric optimization of this PCM jacket will be extended by the addition of cooling tubes.

4.2. *Combined with thermo-chemical heat storage*

Solid-state H_2 storage reactor concept based on thermo-chemical heat storage (THS) can also accomplish the reactor thermal management. This represents an adiabatic system wherein heat is internally and reversibly exchanged between MH and suitable thermo-chemicals. This means that when the metal absorbs hydrogen to release heat, the THS must absorb heat; When the dehydrogenation of MH requires heat absorption, the heat stored in thermo-chemical storage should be provided to MH in time. The main concept involves the integration of exothermic and endothermic reactions to mutually offset each other. As shown in Fig. 9, a and b respectively represent the above two processes. THS is also a reversible process to accomplish the accumulate and release of reaction heat in MH bed [153]. Lutz et al. [154] have demonstrated that the thermal management strategy is completely feasible. At a water vapor pressure of 10 bar, the hydration of MgO can provide a temperature of 300 °C, which is sufficient to promote the dehydrogenation of MgH_2.

There is not much literature on THS for H_2 storage reactor thermal management, but a preliminary comparison can be made. Compared to PCM's thermal management, thermo-chemical energy storage has a higher volumetric storage density and long-term thermal insulation capability. [155]. Besides, the operating temperature of the THS system can also be adjusted, and the temperature difference between the MH bed and the thermo-chemical system can be increased with no change in the hydrogen pressure in the MH [50,154]. Additionally, the cost of using $Mg(OH)_2$ for MgH_2 tanks is only \$20–30, which is significantly lower than using PCM (i.e. \$250–375). Bhouri et al. [153] and Lutz et al. [154] investigated the thermal management performance for MgH_2 MH tanks using THS (Mg $(OH)_2$) and compared the PCM and TES thermal management schemes. The results showed that the required $Mg(OH)_2$ mass was reduced by a factor of four when compared to MH reactor thermal management using PCM, and the authors suggest that, as with PCM, increasing the effective thermal conductivity of the system can reduce the hydrogen uptake and dehydrogenation time [50].

4.3. *Metal hydride heat pump*

With the exception of the above thermal management measures, as shown in Fig. 9c, the thermal management of MH heat pump is also an effective strategy [156,157]. Generally, the reversible thermal effect of alloy hydrogenation and dehydrogenation will be used to promote energy conversion equipment called MH heat pumps. For example, Park et al. [158,159] first conducted extensive research and investigation on the operating characteristics of the compressor-driven MH heat pump. In this work, it was found that the optimal operating conditions of the system were as follows: the cycle time was 2.6 min, the flow rate was 11 m^3/min, and the ambient temperature was 24 °C. Under these conditions, the maximum cooling capacity of the system is 1050 kJ. Muthukumar et al. [160] demonstrated that for a given range operating parameters, the crossed Van't Hoff single-stage metal-based hydride heat pump (SS-MHHP) yields about 60% higher COPs than the conventional SS-MHHP. Tao et al. [161] proposed a novel MH heat pump

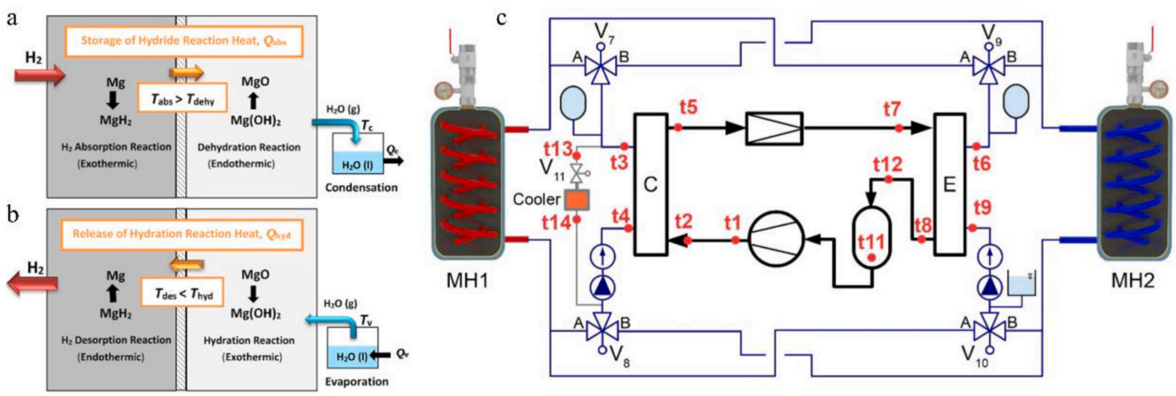

Fig. 9. Operating principle of magnesium hydride reactor using magnesium hydroxide as heat storage media: (a) Absorption of H_2/Dehydration of $Mg(OH)_2$/Condensation of H_2O [153]; (b) Desorption of H_2/Hydration of MgO/Evaporation of H_2O [153]; (c) Diagram of an MH compressor with a heat pump [157,158].

system driven by an electrochemical compressor. Based on this, the relevant thermodynamic models are built to predict the system performance under various parameters. The simulation results show that the compression efficiency of electrochemical compressor has a great influence on the performance of the system, and the system is suitable for refrigeration applications with the lowest efficiency of mechanical compressor below 200 W. Sharma et al. [162] using different combinations of AB_5-type metal hydrides, systematically studied the thermal cycles of four alloy for simultaneous cooling, heat pump and heat conversion. The study shows that the system for the development of metal hydride-based thermodynamic system improves cycle efficiency as well as specific alloy output and facilitates extensive operating temperature range.

4.4. Development tendency of thermal management

The thermal management of MH reactors using PCM and THS has been fully discussed, respectively. Currently, just like THS and the actual PCM material work. At the same time, metal materials or fins can also be added to PCM to reduce the degree of supercooling of PCM and improve the performance of PCM, so that it is very likely that the outer layer of the outer wall of MH bed is likely to grow fins and PCM [163,164]. Similarly, THS and Heat pump also needs to increase the thermal conductivity to speed up the reaction rate. If the algorithm can be used to optimize the scheme of increasing thermal conductivity, it will greatly improve the energy density of thermal management system, which is also an inevitable trend of future development.

5. Hydrogen storage performance (HSP) indicators

To improve the reaction kinetics of metal MH alloys, heat transfer enhancement components need to be added into the MH reactor to enhance the removal/input of reaction heat during hydrogen absorption/desorption. Researchers have proposed lots of novel MH reactors and done a lot of work on the optimization of reactor parameters, as described in Eq. (3) in Section 2. At the same time, when designing enhanced heat transfer in hydride reactors, researchers should also consider the effects on the uptake/discharge rate, reactor mass and volume hydrogen storage density, which are rarely considered in existing studies. In addition, when comparing the performance of hydride reactors with different heat transfer forms, it is difficult to control that the mass and volume of the heat transfer system are the same, so the performance comparison is not on the same basis. One of the main reasons for this situation is the lack of a comprehensive performance indicator that can take into account the hydrogen storage rate, mass and volume storage density of the reactor, because it is inconvenient to use two independent indexes of hydrogen storage rate and hydrogen storage density to guide the heat transfer enhancement design of the reactor at the same time. Therefore, it is more important to establish a unified hydrogen storage performance index. Yang et al. [165–167] put forward several indicators to evaluate reactor performance, such as effective thermal conductivity and reaction rate. However, the existing performance indexes cannot fully reflect the performance of the reactor, especially the heat transfer characteristics, which directly determine the hydrogenation and dehydrogenation rates of MH. As shown in Fig. 10, Meng et al. [168] proposed a evaluation index *Kc* (kg•s^{-1}). This indicator is determined not only by the properties of the alloy, but also by the heat transfer parameters of the reactor, such as heat transfer area, convective heat transfer coefficient, fluid mass flow rate, etc. Under the same working conditions, the higher Kc means a higher reaction rate and better heat transfer performance of MH reactors. Because the higher the Kc, the higher the heat transfer efficiency of the reactor, and the higher the reaction rate. Through numerical simulation and comparative study, the superiority of the new index is proved.

In summary, the current studies about heat transfer design in MH reactors focus on the improvement of hydrogen rate, and pay little attention to the impact on reactor mass hydrogen storage density and volume hydrogen storage density, which have significant effects on the hydrogen storage performance of reactors. In addition, when comparing the performance of hydride reactors with different heat transfer configurations, it is difficult to control the same mass and volume of the heat transfer system, so the performance comparison is not under the same standard. One of the main reasons for this situation is the lack of comprehensive performance evaluation indexes that can take into account the hydrogen storage rate, mass storage density and volume storage density of the reactor, because it is inconvenient to use two independent indexes of hydrogen storage rate and hydrogen storage density to guide the heat transfer enhancement design of the reactor at the same time. It is worth to note that Bai et al. [169] specially further studied the main performance evaluation indexes of MH reactors. A comprehensive hydrogen storage performance (*CHSP*) indicator weighing hydrogen storage rate enhancement and hydrogen storage density loss was constructed, and the weight factors between the two items were determined based on the optimization results. This index mainly considers the effects of hydrogen storage rate, hydrogen storage capacity, volume storage density and mass storage density, and can be used to evaluate the comprehensive hydrogen storage performance of MH reactors. At the same time, this index can also be used to compare the comprehensive performance of hydride reactors with different heat transfer enhancement methods. Wang et al. [170] also proposed an ideal hydrogen storage efficiency based on hydrogen storage time (η_a,η_d), and proposed for the first time a dimensionless index to evaluate the hydrogen storage efficiency of MH reactors, making it possible to compare the hydrogen storage efficiency of different hydrides.

Above all, Kc, CHSP and η_a all reflect partial hydrogen storage properties of MH reactors. Although the evaluation indicators are

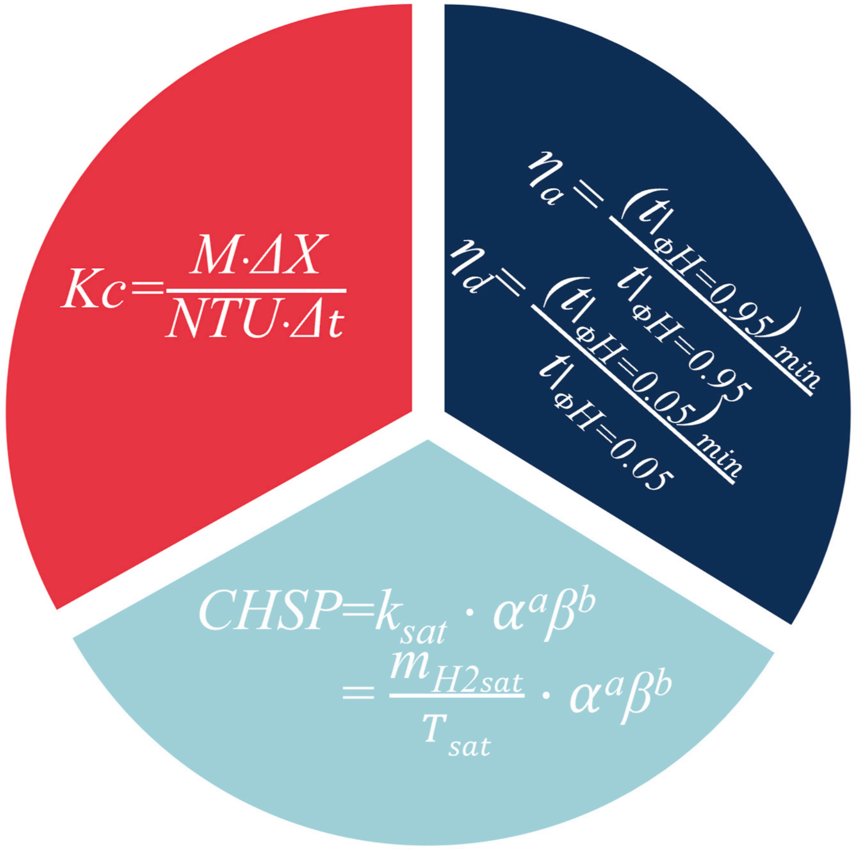

Fig. 10. Different hydrogen storage performance indicators for MH reactors [168–170].

gradually developed, they are still lacking extensive verification, and the comprehensive hydrogen storage performance indicators for the entire MH reactor have not been proposed at present. Therefore, in order to develop a universal standard for the correct evaluation of the hydrogen storage performance of MH reactors, it is necessary for researchers to propose a new indicator that can comprehensively evaluate the comprehensive hydrogen storage performance of MH reactors and continue to verify it. Of course, the new standard must cover the effective thermal conductivity, heat transfer area, convective heat transfer coefficient, fluid mass flow rate, reaction rate, hydrogen storage rate, hydrogen storage capacity, volume hydrogen storage density, mass hydrogen storage density and so on, and also need to have general practicability, not only for individual MH reactors.

6. Conclusion and perspective

By now, in the research of thermal design of MH reactor, whether the chamber geometry, fin shape, cooling tube size or the filling of high thermal conductivity materials are mainly based on the preset structure design, which leads to the limited space or limited material utilization cannot play the best effect, so the partial structure of the MH reaction bed was optimized to obtain the best heat transfer effect and the best reaction rate. During the thermal management of MH reactor, the three thermal management methods mentioned in the review point to a key parameter, thermal conductivity, which will be beneficial to the improvement of the thermal management capability of the system. For the above parts, can be summarized as follows:

In this review, the applications of intelligent algorithms for optimizing the parameters of heat transfer design and thermal management of MH reactors have been sufficiently reviewed. It was found that the addition of high thermal conductivity materials, cooling tube arrangement, and other aspects can be used with intelligent algorithms such as GA, ML, and so on. The chamber and fin shape could be designed with topology optimization to achieve the best efficiency of the system and control the cost at a minimum level. The traditional MH reaction bed research will usher in great changes and development due to the addition of these emerging technologies, which will be unprecedented, and we believe that this will be a development trend and prospect of MH reaction bed structure optimization. On this basis, researchers must also seriously consider the universal indicator that can correctly evaluate the hydrogen storage performance of MH reactors. Only with correct and universally applicable indicator can MH reactors be further developed and better coupled with FC, THS and other system, and thus better serve human beings.

CRediT authorship contribution statement

Ju-Wen Su: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Formal analysis, Conceptualization. **Xin-Yuan Tang:** Writing – review & editing, Writing – original draft, Investigation, Data curation. **Xiao-Shuai Bai:** Writing – review & editing, Writing – original draft, Investigation, Data curation. **Wei-Wei Yang:** Writing – review & editing, Writing – original draft, Supervision, Resources, Funding acquisition. **Jian-Fei Zhang:** Writing – review & editing, Investigation, Data curation. **Zhi-Guo Qu:** Writing – review & editing, Supervision, Investigation, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This work was financially sponsored by the National Key Research and Development Program of China (2023YFB4005404), and Project of Shaanxi Innovative Talent Promotion Plan-Technology Innovation Team (No. 2024RS-CXTD-35).

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